- The FASTQ format

```
@DHKW5DQ1:218:C14L8ACXX:7:2109:14377:51867 1:N:0:GCATGGCT
AACCATAAAACTCACGAATATTTCTAATCGGTTTTAGGATCTCGTGTAGGCTATATTTCAG
+
CCCFFHHFCCCFFFFFGHHHHJJGIGIJJJJFHIIGJIGIJJJJFGHIGGIJIJIIHHGIH
@DHKW5DQ1:218:C14L8ACXX:7:2212:8965:2922 2:N:0:GCATGGCT
AACCATAAACTGACGAATATTTCTAATCGGTTTTAGGATCTCGTGTAGGCTATATTTCAGA
+
CCCFFHHFCCCFFFFFHHHHHJJJJJJJJIJHGIJIJJEGIJIJHGGGGGIIJJIJIJIJIG
@DHKW5DQ1:218:C14L8ACXX:7:1116:16205:46371 1:N:0:GCATGGCT
AACCATAAGGCTAATATTTCTAATCGGTTTTAGGATCTCGTTGCTGTGTTTAAAATTTCAG
teresa@teresa-N551JM[data] head -n 20 reads.fastq
@DHKW5DQ1:218:C14L8ACXX:7:2109:14377:51867 1:N:0:GCATGGCT
AACCATAAAACTCACGAATATTTCTAATCGGTTTTAGGATCTCGTGTAGGCTATATTTCAG
+
CCCFFHHFCCCFFFFFGHHHHJJGIGIJJJJFHIIGJIGIJJJJFGHIGGIJIJIIHHGIH
@DHKW5DQ1:218:C14L8ACXX:7:2212:8965:2922 2:N:0:GCATGGCT
AACCATAAACTGACGAATATTTCTAATCGGTTTTAGGATCTCGTGTAGGCTATATTTCAGA
+
CCCFFHHFCCCFFFFFHHHHHJJJJJJJJIJHGIJIJJEGIJIJHGGGGGIIJJIJIJIJIG
@DHKW5DQ1:218:C14L8ACXX:7:1116:16205:46371 1:N:0:GCATGGCT
AACCATAAGGCTAATATTTCTAATCGGTTTTAGGATCTCGTTGCTGTGTTTAAAATTTCAG
+
CCCFFHHF?@;BD32:CFDDFFBAFE6+<CF@33CH<<4C?@D8C<00BBCBDC>DCF499
```

• The FASTQ format

- A FASTQ file is a text file format used to store sequenced reads
- The FASTQ file format includes the following four lines for each sequence:

- Line 1: begins with a '@' character and is followed by a sequence identifier and an optional description (like a FASTA header line).
- Line 2: holds the raw sequence.
- Line 3: begins with a '+' character. The mandatory '+' sign can be optionally followed by the same sequence identifier (and description, if any) as in line 1 after the '@' sign.
- Line 4: encodes the quality values for the sequence in Line 2, and must contain the same number of symbols as letters in the sequence. A single FASTQ entry would look like this:

```
@SEQUENCE UNIQUE HEADER XXX

GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT
+
!''*((((***+))%%%++)(%%%%).1***-+*''))**55CCF¿¿¿¿¿¿CCCCCCC65
```

- Quality scores of a sequenced read (Line #4 in a FASTQ record)

- The quality (confidence level) associated with each letter (A,T,C,G) in a sequence is indicated by an ASCII (*) symbol. ASCII includes definitions for 128 characters.

- The advantage of using a one-symbol quality encoding is to have a one-to-one matching between a given sequence letter and its quality score, thus the length of the sequence read (line #2 in FASTQ record) is equal to the length of the quality scores (line #4).

- The following 93 ASCII characters (from the 33rd to the 126th) are used to encoding the quality of a sequence, listed from lowest to highest quality:

- !"#$%&'()*+,-./0123456789:;<=>@ABCDEFGHIJKLMNOPQ
  - RSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~

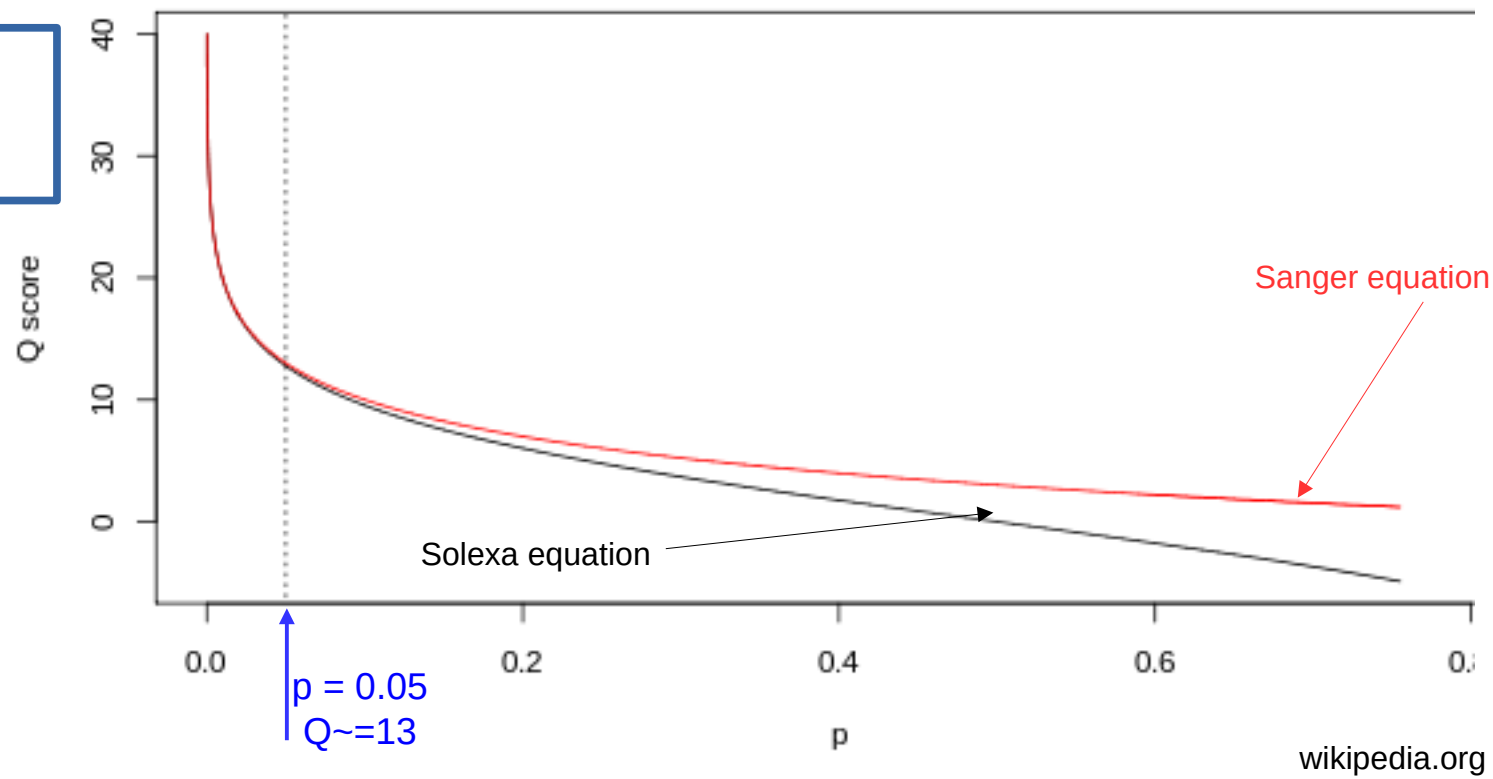
- Quality scores of a sequenced read (Line #4 in a FASTQ record)

A quality value (Q) is an integer mapping of the probability that the corresponding base call is incorrect (p).

Two standard Sanger (*) equation to assess reliability of a base call, otherwise known as Phred (**) quality score, is:

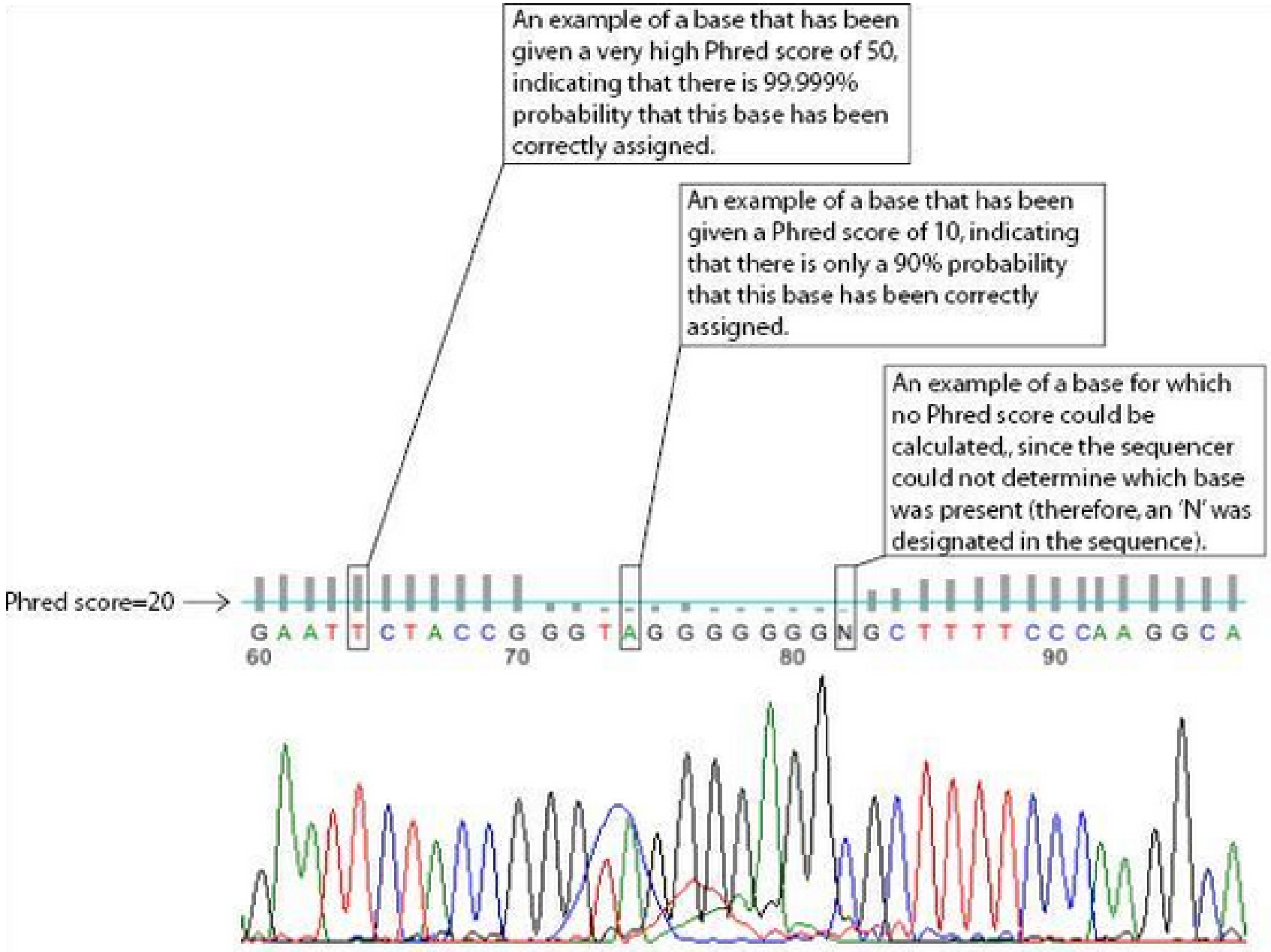
$$Q = -10 * \log10 (p)$$

Prob. of incorrect base call	Phred quality Score	Base call accuracy
1 in 10	10	90%
1 in 100	20	99%
1 in 1000	30	99.9%
1 in 10000	40	99.99%
1 in 100000	50	99.999%



(*) from *Wellcome Trust Sanger Institute*, the sequencing institute that firstly introduced the FastQ format.
(**) after the name of the software that firstly intriduced this quality measure.

- Quality scores of a sequenced read (Line #4 in a FASTQ record)



FASTQ format

```
@SRR047027.1 GFTKDII01EMEVN length=68
CACGGAATTAGCCGGTGCTTATTCTTCAGGTACCGTCATCAGCTGTCGATATTAGCAACAGCCTTTTC
+SRR047027.1 GFTKDII01EMEVN length=68
FFFGGFFFFEEDDGFFFFFFHFFFFFF@??FFCCDFFFFFFFFFFFFDDDFFFDBAA>?<:22..,,1
@SRR047027.3 GFTKDII01ETSLN length=54
CACGTAGTTAGCCGTGGCTTTCTGGTTAGATACCGTCAAGGCACACAGGGAGTA
+SRR047027.3 GFTKDII01ETSLN length=54
IIIIIIIIIIIIIIIIIHHHIIIIIIIIIIIIIIIHHHIIIIIIFA999;7>8
@SRR047027.5 GFTKDII01ANQX8 length=44
CACGTATTTAGCCGTCACTTCTTCTGTTGGTACCGTCATTTTTT
+SRR047027.5 GFTKDII01ANQX8 length=44
IIIIIIHHHIIIIIIIIIIIIIIIIIIIFFFFBA>??=-,,,,,
@SRR047027.7 GFTKDII01D721L length=215
```

vs.

FASTA format

```
>9181024
AGCCGCGCCGAATTCGTGGGCACGCTGGTCGAAGGCATCGTCAGCTTCACGCCCGGCGGGCGTTTCCTGGCGGCCAACCGCAGCGCGCAGTTCCAGCTGGGG
>9181029
TGTTCGGCCGATTGGTGATTCGAGATCAGCATCAAGCTCTCTCGAAACGGGATCCCCTTCTCGTCATGGGAGATGAGCCTAGCGACAG
>9181034
TGCGCAATTGCCGTGGCAGCAGATAGGCCAGGATCACCCGCGCCGTGGCGCTGTGGAACAGGTCCATGGGCGTGCCGCGCCCATAGTTCAGGCG
>9181039
CAGGCCGATACCGAGTATCCAGGCGAAATACCACATGATGTTCTCCTCGTTGCAGTTTGCGATTGCGCGTAATCAATACGCCGTGTGTTCGT
>9181044
CAGCAGCACGCGCGCCACACCGCTGCGGGCCGCGGCACTGGTGGCTTCGTCGCCGCTGTAATAGTCGCCCAGCAGGCGCTGCAG
>9181049
CGATGCAATGAGGCAGGTTATGGAGGCGACGAGTGTGGAGAGGCAGATGAAGAGCTTTGAGGCGGCGAAGGAGAGGGAAAGGGCTGAGAAGGGCGTTCCG
>9181054
TTGAATCATTTATTGACCGCAGAGCAGATTCAGTTGGTGTTATCCGCTGCTACTATGGAGGAACTTAGTCAAATATGCAAAACGTTTCTACGCAATTTG
>9181059
GAAGAGGTGGGCTTTGAGCAGCCGCTTAAAACTGGATGGGGAAGACTATTCCACAATGTTGGGGCTAAAACAGCGAAGGCACGGTCACCTTTAGATTTACAGGTAGAGCATGG
>9181064
CTCGCCGTGATAGATGTTGTCACCCAGAATCAGTGCCGAGTTCTCGCCGTTGATGAAGTCGCGGCCGATGATGAAGGCTTGTGCCAGACCA
>9181069
AGGATTGGCAACTCCTAGTGATCTTAGGAATGCATTGATGGGACCATCCTGGGCATTGTAAAGCACACTCCACACCACAGAAACGGCAACCAAACTG
```

CHALLENGE

- By looking at the full list of ASCII symbols, can you guess why the symbols adopted for quality encoding start at ASCII symbol number 33?

- What are the quality score values corresponding to the first 5 letters of the following sequence?

```
@SRR390728.3 3 length=72
CCAGCCTGGCCAACAGAGTGTTACCCCGTTTTTACTTATTTATTATTATTATTTTGAGACAGAGCATTGGTC
+SRR390728.3 3 length=72
-;;;8;;;;;;;,*;;';-4,44;,:&,1,4'./&19;;;;;;669;;99;;;;;-;3;2;0;+;7442&2/
```

- Are they good or bad scores? To what probability of incorrect base calling do they correspond?

Note: Full ASCII reference table is available here: http://www.oppo.it/tabelle/tabella_ascii.htm